

Armature Reaction →

- * On NL the arm. current is negligible. Consequently there is only main flux in the air gap distributed uniformly in flat topped nature.
- * As the arm. is loaded load current flows in the arm. conductors which produce arm. mmf & arm. flux which is also distributed uniformly throughout the arm. peripheral in the air gap.
- * The arm. flux will take an action on main flux distribution which is called as arm. reaction.

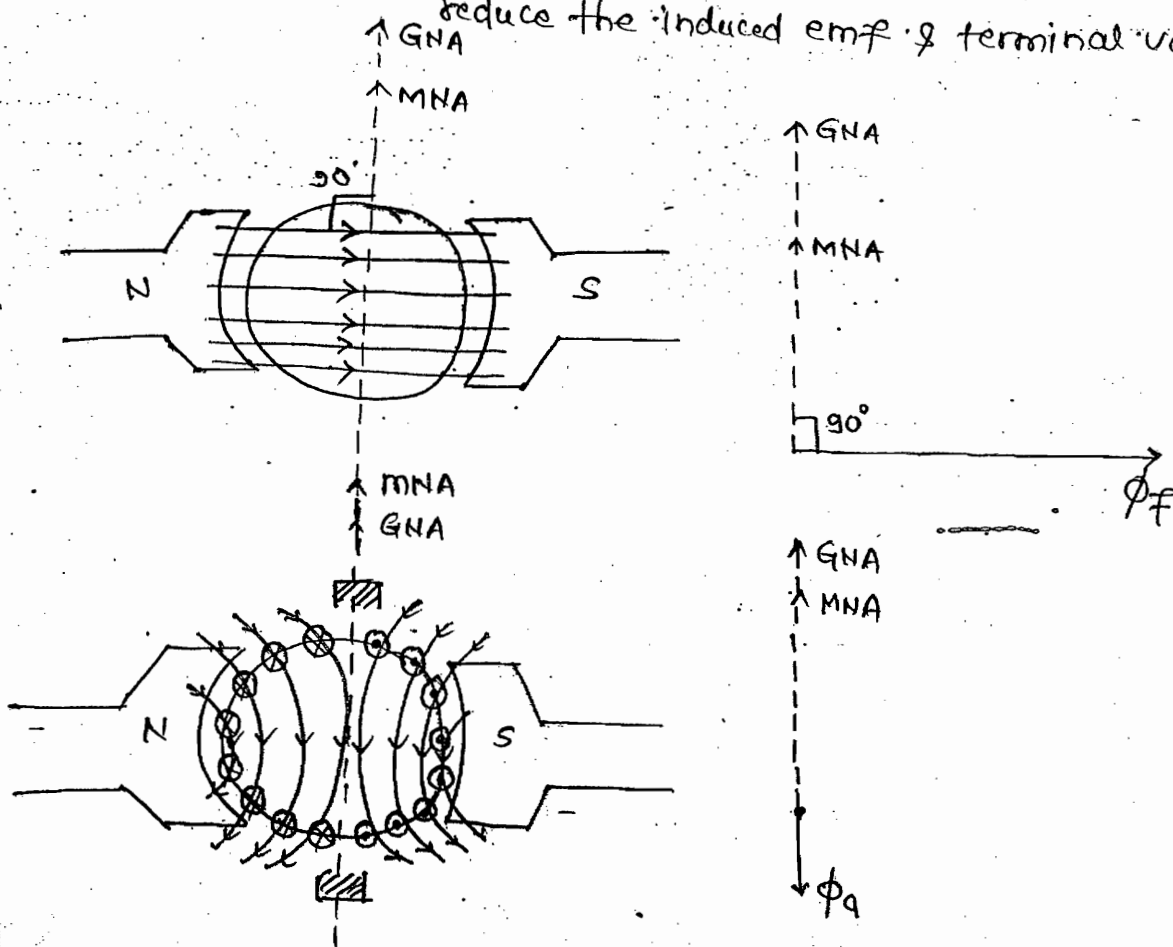
* The effect of the arm. flux on main flux produce :-

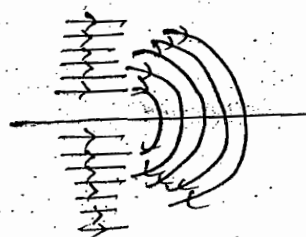
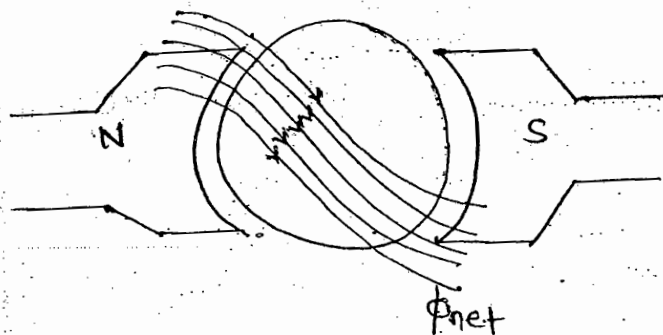
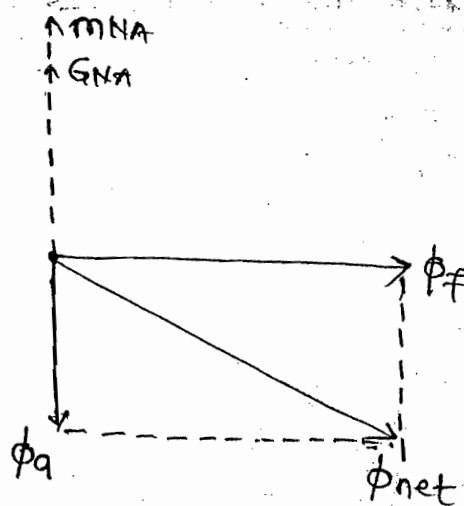
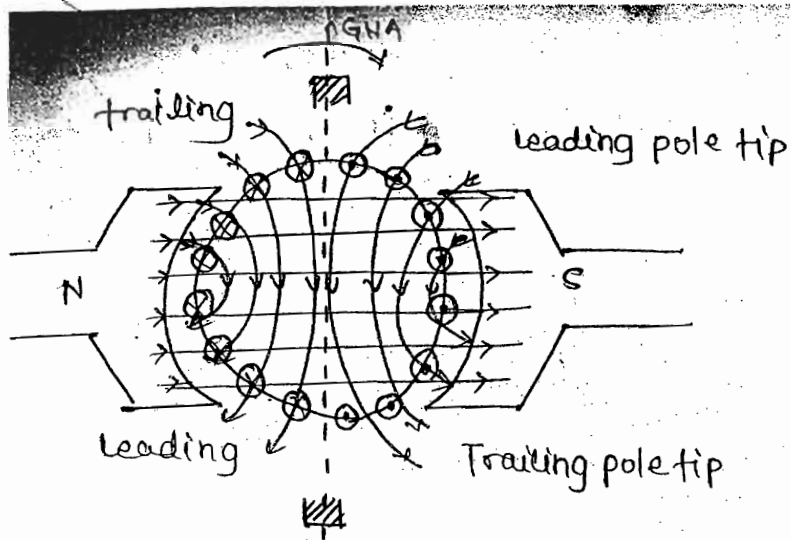
(1.) Cross magnetisation.

(2.) Demagnetisation.

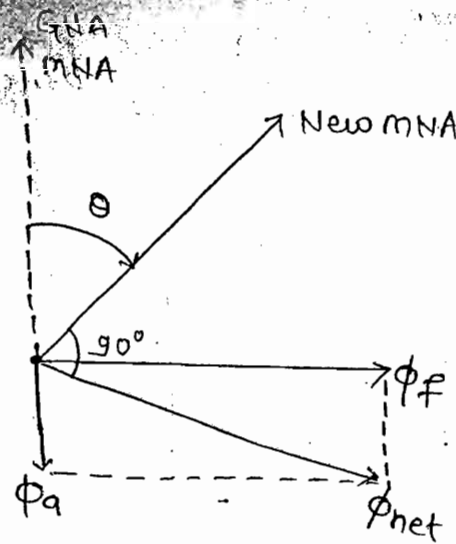
Cross magnetisation → It is distortion of main flux due to which MNA is shifted, & the commutation will not be successful due to sparking.

Demagnetisation → It is Reduction in the main flux which reduce the induced emf & terminal voltage.



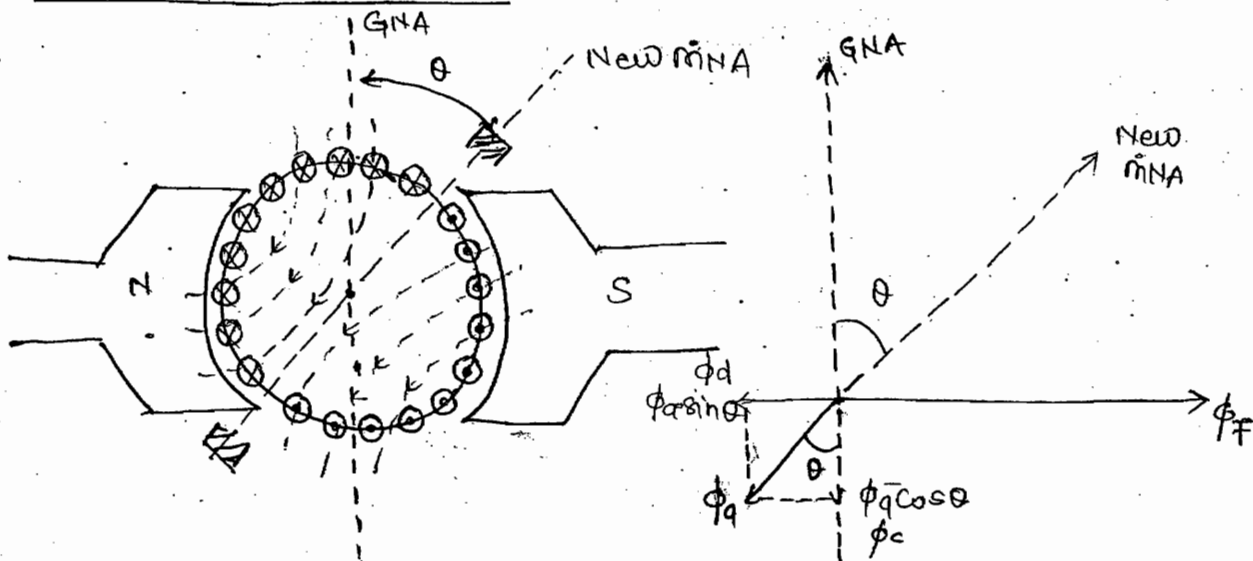


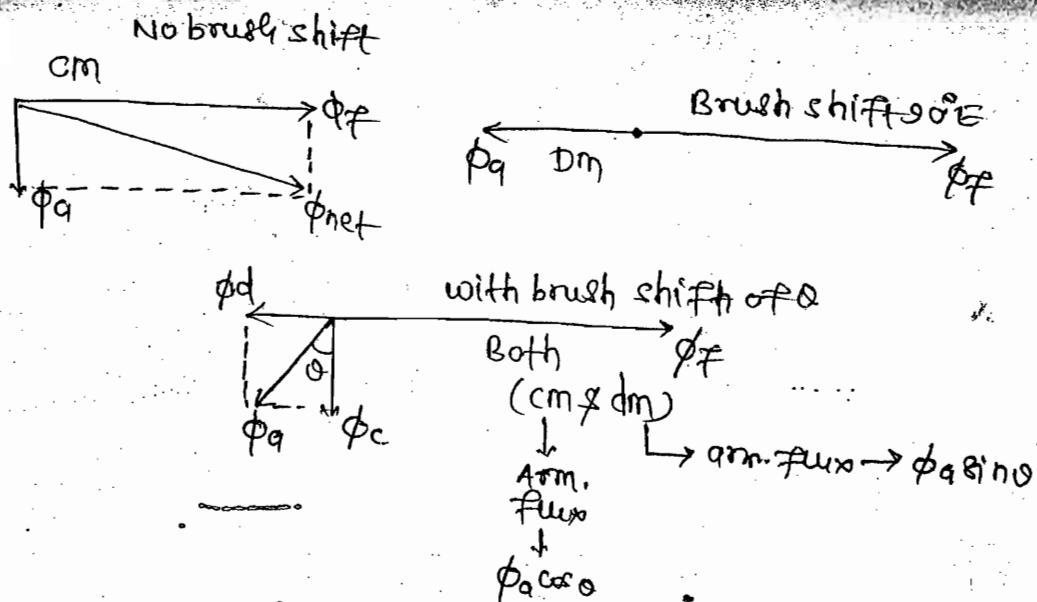
- * Depending on the dirn of rotation the pole tips are named as leading & trailing.
- * The arm. flux under the trailing pole tips of gen^r will increase the flux density as it is in the same dirn. with the main flux.
- * The arm. flux under the leading pole tips will demagnetise the main flux as it is in the opposite dirn.
- * If the amount of magnetisation & amt of Dm are equal then there is no net reduction in main flux but only distortion.
- * Under practical condⁿ the poles get magnetically saturated & consequently the increase in flux density under trailing pole tips is comparatively less than that of decrease in the flux under leading pole tips.



- * Due to arm. flux which is also called as cross flux the main flux is distorted. known as cross magnetised due to which mNA is shifted in the dirn of rotation of genr.
- * In order to improve commutation the brushes also need to be shifted in the dirn of rotation of genr.
- * Due to the affect of brush shift to an angle θ there exist additional demagnetisation.
- * Brush shift is not done generally. It has been replaced with interpole as it is not reliable method & also produce additional demagnetisation.

Affect of brush shift $\rightarrow$





ϕ_c or $\phi_a \cos \theta$ is cross magnetisation of arm. flux

$\phi_d = \phi_a \sin \theta$ is demagnetisation of arm. flux

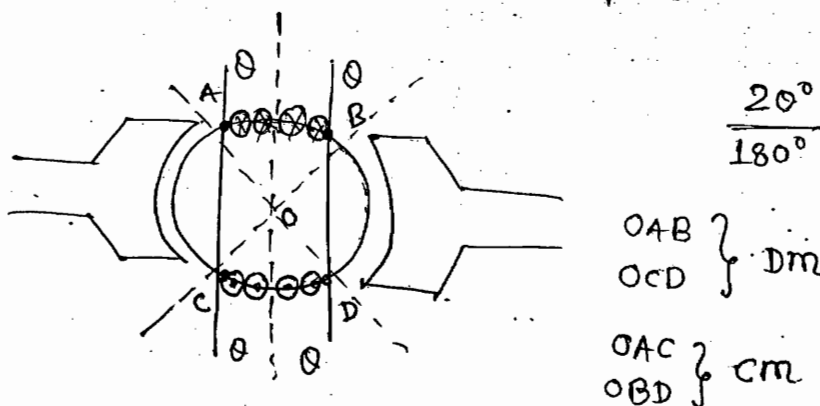
when $\theta = 0$

$\phi_d = 0, \phi_c = \phi_a$ (arm. flux is cm in nature)

when $\theta = 90^\circ$

$\phi_d = \phi_a, \phi_c = 0$ (arm. flux is demagnetisation in nature)

De-magnetising Amp-turns/pole $AT_d/p \rightarrow$



$$\frac{2\theta}{180^\circ} \times \frac{I_a}{A} \times \frac{Z}{2P}$$

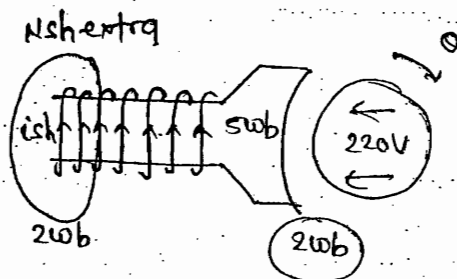
Gross magnetising Amp-turns/pole $\rightarrow$

$$AT_c/p = \frac{180-2\theta}{180^\circ} \times \frac{I_a}{A} \times \frac{Z}{2p}$$

$$\text{Total arm. mmf} = \frac{I_a}{A} \times \frac{Z}{2p}$$

* AT_d/p is representing the arm. flux which produce additional demagnetisation.

* In order to compensate this additional demagnetisation extra amp-turns need to be provided on each pole.



$$N_{sh \text{ extra}} \times I_{sh} = AT_d/p$$

$$N_{sh \text{ extra}} = \frac{AT_d/p}{I_{sh}}$$

$$\frac{AT}{A} = T$$

* The no. of extra turns to be added on each pole of shunt gen^r in order to compensate additional demagnetisation produced by the brush shift is equal to $AT_d/p / I_{sh}$.

* Similarly in a series gen^r

$$N_{se \text{ extra}} = \frac{AT_d/p}{I_a \text{ or } I_{se}}$$

DATE - 07/07/14

* Upper effects of arm. reaction $\rightarrow$

(1) Decrease in efficiency due to increased iron loss $\rightarrow$

The increased flux density under the pole tips will increase iron loss in the core as iron loss is directly proportional to flux density.

(2) Increased maintenance & repair $\rightarrow$

Due to CM Commutation is not successful & there will be unacceptable sparking which damage the brush surface.

(3) Increased design cost $\rightarrow$

* Methods to reduce armature reaction & its effects $\rightarrow$

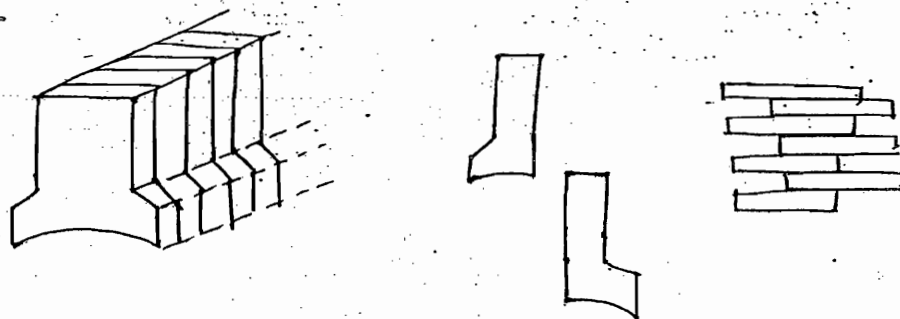
(1) Pole stacking

(2) Pole chamfering

(3) Pole core slotting

(4) Compensating wdg.

(1) Pole stacking $\rightarrow$



* The pole laminations are alternately stacked to introduce air gap under the pole tips.

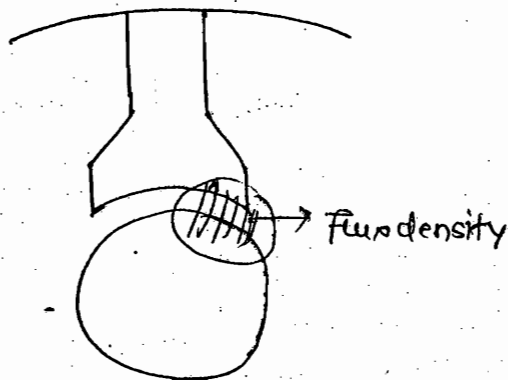
* The increased reluctance will reduce the flux density & reduce the iron loss but the net reluctance in the m/c increased which demands more mmf which increase the size & cost of m/c.

-stacking $\rightarrow$ $\uparrow$ Reluctance

② Pole chamfering →

min^m reluctance at the center & increased reluctance towards the pole tips.

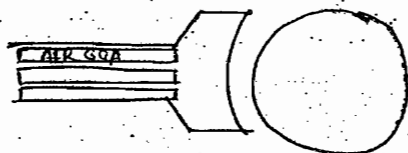
Concept is same as pole stacking.



③ Pole core slotting →

* The pole core contains rectangular slots to introduce air gap to some part of the flux, & reduces it to some extent.

$$\text{Flux} = \frac{\text{MMF}}{\text{Reluctance}}$$



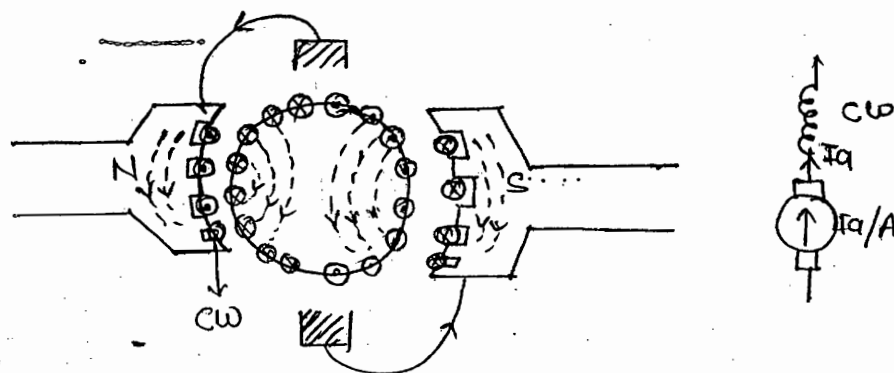
*** (4) Compensating winding →

* In large rating dc m/c operating at varying load condⁿ running at high speeds compensating wdg are essential in order to reduce flashover on the commutator.

* As the load vary arm. current vary & produce a varying flux on the arm. links with arm. cond^r. & produce statically induced emf, which results in circulating currents which interfere with commutation & produce sparking.

* Due to high speeds if the spark spreade it will become a flash over to damage the wdg. Compensating wdg is provided in the pole shoe or pole face by cutting into teeth or slots.

- * It is always connected in series with arm wdg. through brushes
- * In order to automatically neutralise the arm. flux under the pole.



- * The current flowing in the cw under any pole should exactly opposite dirn to the arm cond^r current dirn under the pole. in order to cancel out the arm. flux

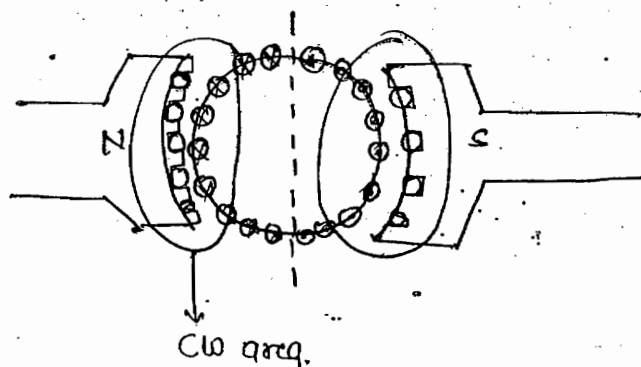
Let z_c be cw cond^r, z be AW cond^r

$$I_a \cdot \frac{z_c}{2} = \frac{I_a}{A} \cdot \frac{z}{2}$$

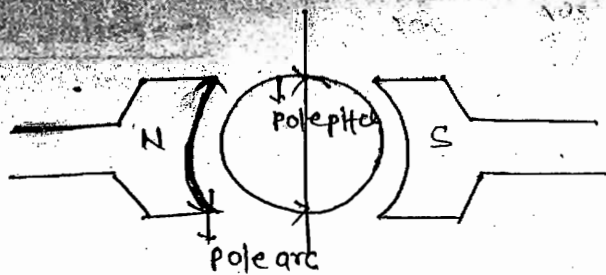
$$z_c = z/A$$

No. of compensating wdg under each pole

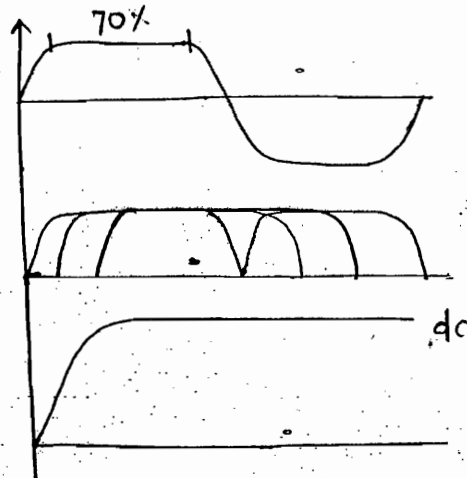
$$z_c = z/A_p$$



Pole area	Factor = 0.7 or 70%
Pole pitch	



* Flux distribution under the pole in dc m/c is Flat topped nature.



$$Z_c = \frac{Z}{A_p} \times \frac{\text{Pole Arc}}{\text{pole pitch}}$$

Because Z_c is placed on pole arc region that's why multiplying with pole arc/pole pitch.

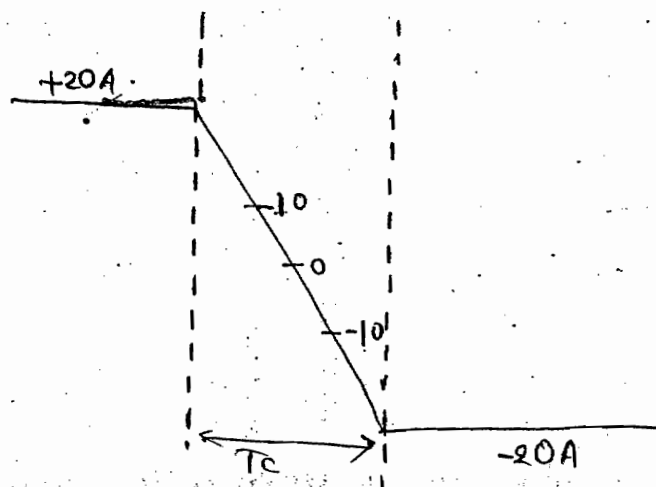
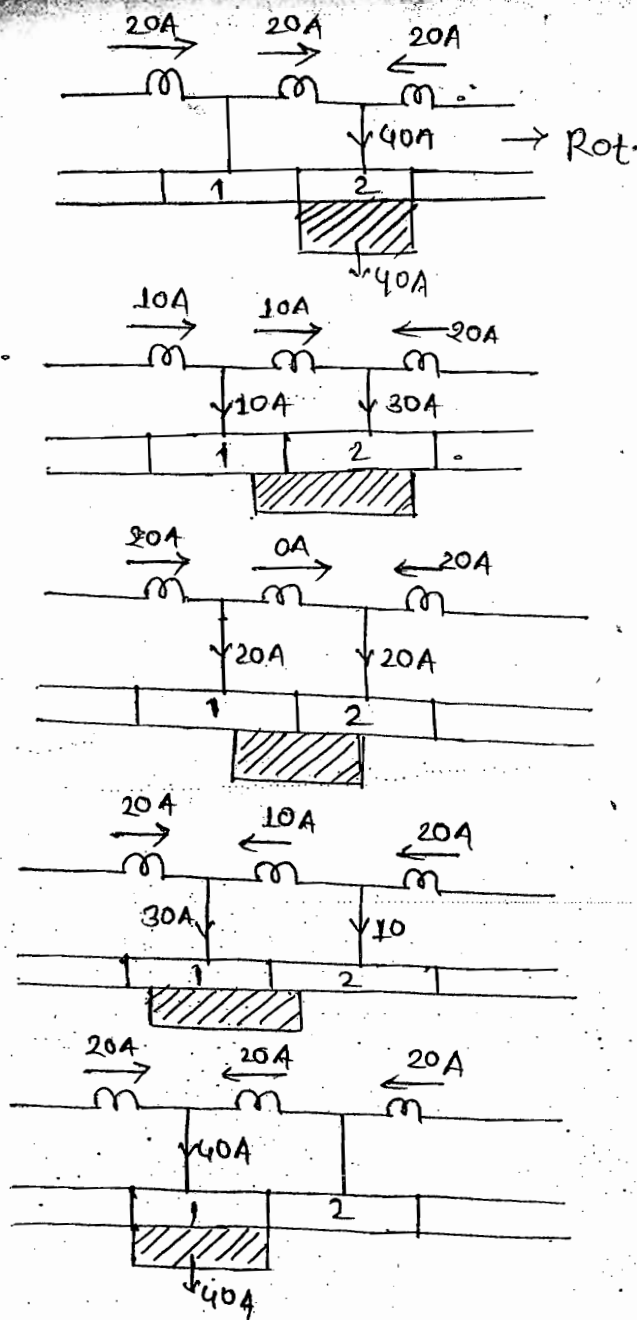
* COMMUTATION → The process of current reversal in the coil when it passes through a brush is known as commutation.

* The time taken for the brush to span from 1 segment to the other is known as commutating time.

* If the current reverses completely within the commutation time in the coil undergoing commutation then commutation is successful also called as linear, ideal or straight line commutation.

* There will be no sparking at the brush.

* If the current doesn't reverse completely within the commutating time in the coil undergoing commutation there will be sparking at the brush & the commutation is unsuccessful known as delayed commutation or non-linear.



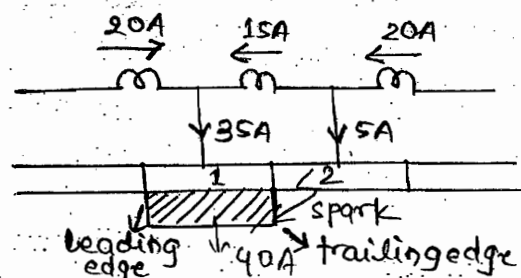
Reactance voltage →

$$e_r = N \frac{d\phi}{dt} = L \frac{di}{dt}$$

$\nearrow 2I$
 $\rightarrow T_c$
 $\rightarrow L_{self}$

* In the coil undergoing commutation there is a total change of current from I_a/A to $-I_a/A$ within commutating time T_c which produce the reactance voltage due to self inductance property of the coil.

* According to Lenz law it will oppose the cause & change in current. Therefore by the end of commutating time T_c the current will not be reversed completely.



* Any unchanged current by the end of commutating time will jump into the brush through spark at the trailing edge of brush.

Methods to improve Commutation →

(1) Resistance Commutation

(2) Voltage Commutation $\begin{cases} \rightarrow \text{Brush shift} \\ \rightarrow * \text{Interpole} \end{cases}$

① Resistance Commutation → * Replacing low resistance Cu brush

with high resistance C brush to improve

commutation by reducing the chances of sparking to some extent.

* C brushes have high resistance compare to Cu.

* Due to its high resistance C brush doesn't encourage sparking at trailing edge & improve commutation.

* The added ad. of c brush are:-

(1) It is not hard material as Cu.

(2) It is self lubricating (polishing) in nature which offers good mech. condⁿ with the comm. surface.

(3) If any spark occurs it will get less damaged than Cu.

disad. →

① more brush contact drop.

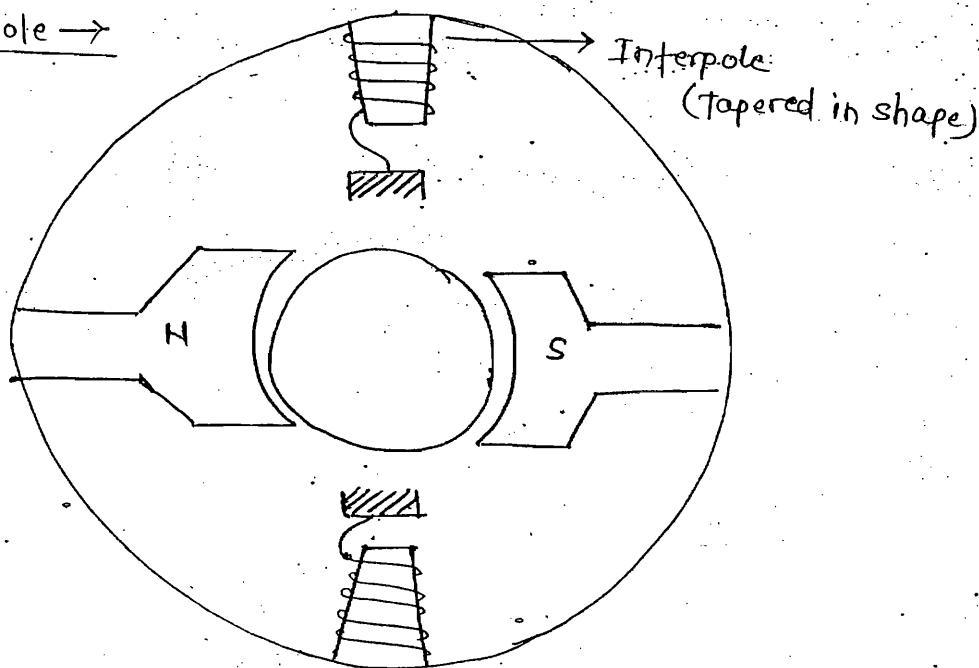
② Low current density, requires larger brush.

* Brush shift → * It is the 1st method of improving comm. which is complicated & not reliable because m.m.f. changes with loads continuously & after the design brushes can't be changed.

* Due to brush shifting there is additional Dm.

* It is not ~~turn~~ done after the invention of interpole.

* Interpole →



* Interpoles are small poles compare to main poles placed in the interpolar region between the main poles on the yoke.

* There are also electromagnets with interpole wdg which is connected in series with arm. wdg through brushes in order to have automatic neutralisation of arm. flux in the interpolar region.

* It performs 2 funⁿ:-

(1) Produce a counter flux on the coil undergoing commutation to nullify the reactance voltage (2) It also remove the inequality in the flux densities on the top & bottom region of arm.

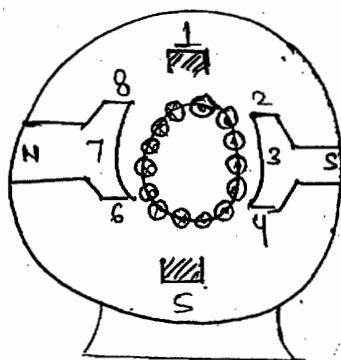
* The interpoles are tapered in shape with comparatively more air gap in order to avoid easy saturation as the load current flows in the interpole wdg. The no. of turns on the interpole are calculated acc to $\text{cm} \cdot \text{amp turns} / \text{pole}$.

* More specifically it is the amp turns in $\frac{1}{2}$ of arm. in the interpolar region, & some additional turns are also required in order to produce reactance voltage.

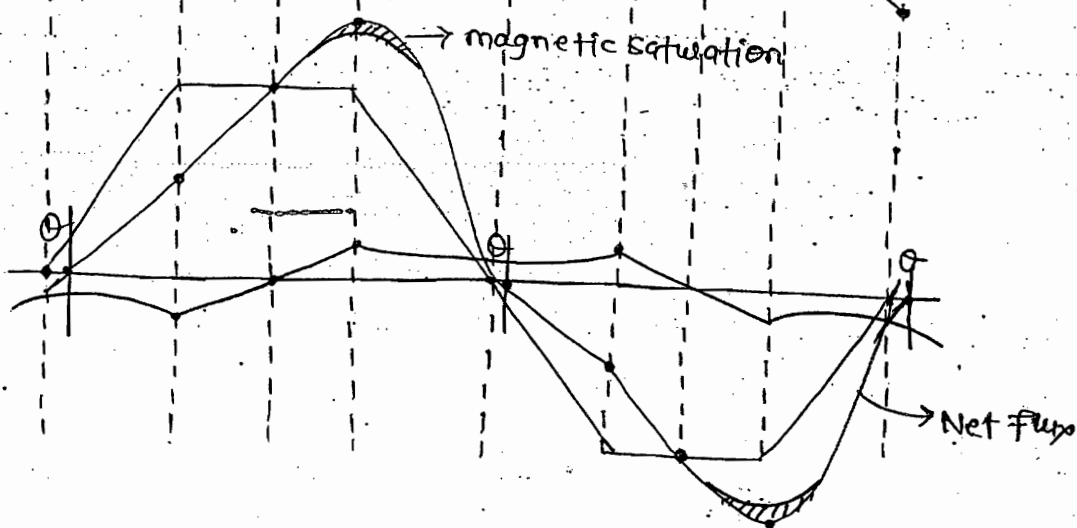
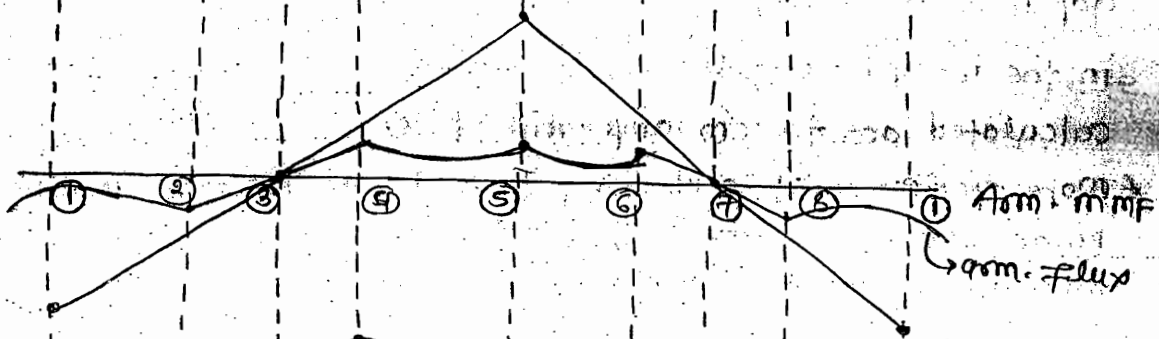
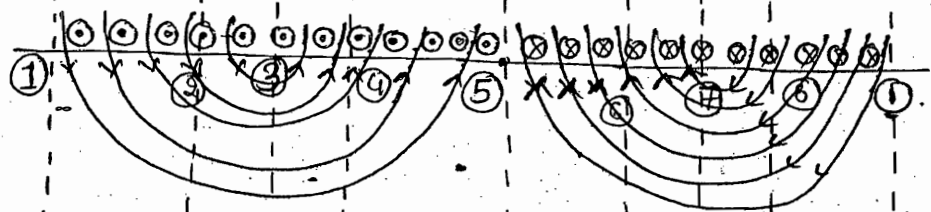
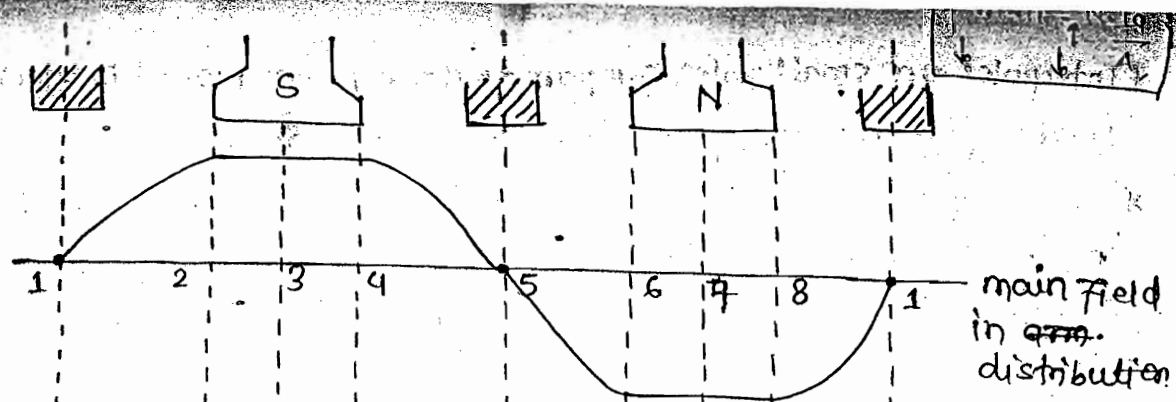
CLW $\rightarrow$ Reduce arm. reaction

IPW $\rightarrow$ Improve Commutation

Interpole are directly improve Comm. & CLW indirectly improve Comm.



	turns (or) amp F	
1-3	↓	-ve
3-5	↑	+ve
5-7	↓	+ve
7-1	↑	-ve
3-1	↑	-ve



(Reluctance)

* In the m/c if air gap is uniform then the arm. mmf & arm. flux follow the same shape but magnitude may vary.

$$\text{Armature Flux} = \frac{\text{Arm. mmf}}{\text{Reluctance (Air gap)}}$$

	Reluctance	
1-2	↓	-ve
2-3	↓ min ^m	-ve
3-4	↑	+ve
4-5	↑ max ^m	+ve
5-6	↓	+ve
6-7	↓ min ^m	+ve
7-8	↑	-ve
8-1	↑ max ^m	-ve

* For drawing arm. flux ^{mmf} arm. flux is taken as ref.

- (1) Main field flux distribution is having flat topped shape. or trapezoidal.
- (2) Arm. mmf is triangular in shape increasing or directed towards the brush axis.
- (3) Net. flux due to arm. reaction is having peaky. (If not given peaky then Δ)
- (4) Arm. flux is saddle shape (If not given then prefer Δ).

*** The main flux & arm. flux are always 90° E wrt each other (quadrature or ⊥) / orthogonal.

* Without arm. flux. the main flux is exactly 0 along GNA.

* With arm. flux the neutral (MNA) has been shifted in the dirn of rotation of genr. ~~to~~ ~~away~~

* To improve comm. the brushes need to be given a forward shift.